

Physics 7B Final Q1 Solution

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- (a) Given $\frac{P}{T^2} = \text{constant}$, we start from the ideal gas law $PV = nRT$, or $T = \frac{PV}{nR}$. Substituting into the process relation:

$$\frac{P}{T^2} = \frac{P(nR)^2}{(PV)^2} = \frac{(nR)^2}{PV^2} \Rightarrow PV^2 = \text{constant} = C \quad (1)$$

Using $P_A V_A^2 = P_B V_B^2$ and $P_B = 4P_A$, we solve:

$$P_A V_A^2 = 4P_A V_B^2 \Rightarrow \boxed{V_B = \frac{V_A}{2}} \quad (2)$$

- (b) Since $PV^2 = \text{constant}$, the curve is steeper than an isotherm. The curve drops sharply in volume and rises in pressure.

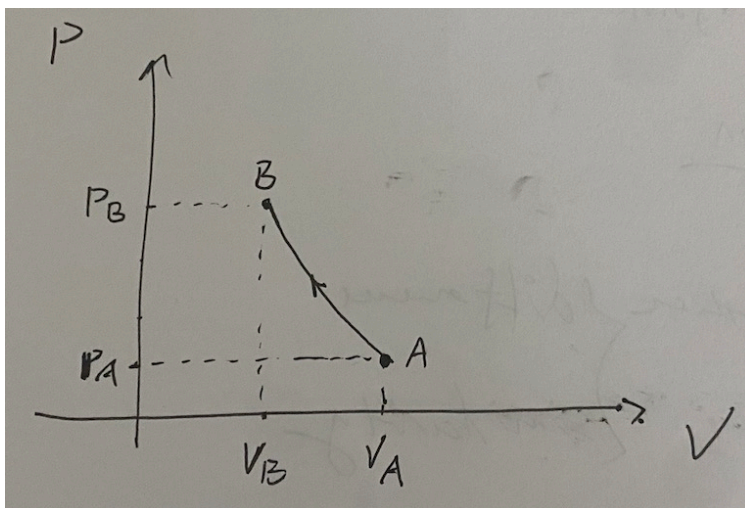


Figure 1: "P-V Diagram"

- (c) The work done **by the gas** is:

$$W = \int_{V_A}^{V_B} P dV = \int_{V_A}^{V_B} \frac{C}{V^2} dV = -\frac{C}{V} \Big|_{V_A}^{V_B} \quad (3)$$

Where C is the constant coefficient, $C = P_A V_A^2$, and $V_B = \frac{V_A}{2}$:

$$W = -P_A V_A^2 \left(\frac{2}{V_A} - \frac{1}{V_A} \right) = \boxed{-P_A V_A} \quad (4)$$

The work done is negative because the volume decreases from state A to state B . Therefore, the gas does negative work.

(d) Using the formula to calculate entropy change for a reversible system:

$$\Delta S = \int_A^B \frac{dQ_{\text{rev}}}{T} \quad (5)$$

Using the first law of thermodynamics to express dQ

$$dQ = dU + dW \quad (6)$$

For a monoatomic ideal gas, the internal energy is:

$$dU = \frac{3}{2}nRdT \implies dQ = \frac{3}{2}nRdT + P dV \quad (7)$$

So the entropy change becomes:

$$\Delta S = \int \frac{dQ}{T} = \int \left(\frac{\frac{3}{2}nR dT}{T} + \frac{P dV}{T} \right) \quad (8)$$

We now evaluate each term separately.

1. Temperature contribution:

$$\int_{T_A}^{T_B} \frac{\frac{3}{2}nR}{T} dT = \frac{3}{2}nR \ln \left(\frac{T_B}{T_A} \right) \quad (9)$$

2. Volume contribution: Using the ideal gas law:

$$\frac{P}{T} = \frac{nR}{V} \implies \int_{V_A}^{V_B} \frac{P}{T} dV = \int_{V_A}^{V_B} \frac{nR}{V} dV = nR \ln \left(\frac{V_B}{V_A} \right) \quad (10)$$

Total entropy change:

$$\Delta S = \frac{3}{2}nR \ln \left(\frac{T_B}{T_A} \right) + nR \ln \left(\frac{V_B}{V_A} \right) \quad (11)$$

From the process:

$$\frac{T_B}{T_A} = \frac{P_B V_B}{P_A V_A} = \frac{4P_A \cdot \frac{V_A}{2}}{P_A V_A} = 2, \quad \frac{V_B}{V_A} = \frac{1}{2} \quad (12)$$

Therefore:

$$\Delta S = \frac{3}{2}nR \ln(2) + nR \ln \left(\frac{1}{2} \right) = \frac{3}{2}nR \ln(2) - nR \ln(2) = \frac{1}{2}nR \ln(2) \quad (13)$$

$$\boxed{\Delta S = \frac{1}{2}nR \ln(2) > 0} \quad (14)$$

Interpretation: The entropy of the gas increases: $\Delta S = \frac{1}{2}nR \ln(2) > 0$. This does **not** violate the second law of thermodynamics.

In a reversible process, the total entropy change of the system and its environment must be zero. The positive entropy change of the gas is exactly canceled by an entropy decrease in the environment (due to heat loss). Therefore:

$$\Delta S_{\text{universe}} = \Delta S_{\text{gas}} + \Delta S_{\text{env}} = 0$$

The result is fully consistent with the second law.

Final Q1 Grading Rubric (Total: 25 points)

(a) **Final Volume V_B** [6 points]

- Starts with correct application of the ideal gas law and process equation — 1 pt
- Substitutes $T = \frac{PV}{nR}$ into $\frac{P}{T^2} = \text{const}$ — 1 pt
- Derives $PV^2 = \text{const}$ — 2 pts
- Applies relation $P_A V_A^2 = P_B V_B^2$ and solves for V_B — 1 pt
- Correct final answer $V_B = \frac{V_A}{2}$ — 1 pt

(b) **P-V Diagram** [4 points]

- Labeled axes: 1 pt
- Correct start/end states: 1 pt
- Correct shape of curve: 1 pt
- Correct direction: 1 pt

(c) **Work Done** [7 points]

- $W = \int P dV$ Integral set up: 2 pts
- Substitution of $P = C/V^2$: 2 pts
- Correct Magnitude of work: $|W| = P_A V_A$: 1 pt
- Correct Negative sign: 1 pt
- Correct interpretation of negative sign(compression): 1 pt

(d) **Entropy Change and Second Law** [8 points]

- Starts from correct entropy definition: $\Delta S = \int \frac{dQ_{\text{rev}}}{T}$ — 1 pt
- Applies the first law: $dQ = dU + PdV$, and uses $dU = \frac{3}{2}nRdT$ for monoatomic ideal gas — 2 pts
- Correctly splits the integral:

$$\Delta S = \int \frac{\frac{3}{2}nR}{T} dT + \int \frac{P}{T} dV$$

and substitutes $\frac{P}{T} = \frac{nR}{V}$ — 2 pts

- Computes both integrals:

$$\Delta S = \frac{3}{2}nR \ln \left(\frac{T_B}{T_A} \right) + nR \ln \left(\frac{V_B}{V_A} \right)$$

with correct evaluation using:

$$\frac{T_B}{T_A} = 2, \quad \frac{V_B}{V_A} = \frac{1}{2} \Rightarrow \Delta S = \frac{1}{2}nR \ln 2$$

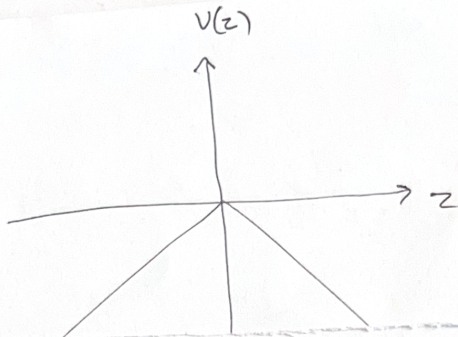
— 2 pts

- Correct interpretation, no violation of the second law — 1 pt

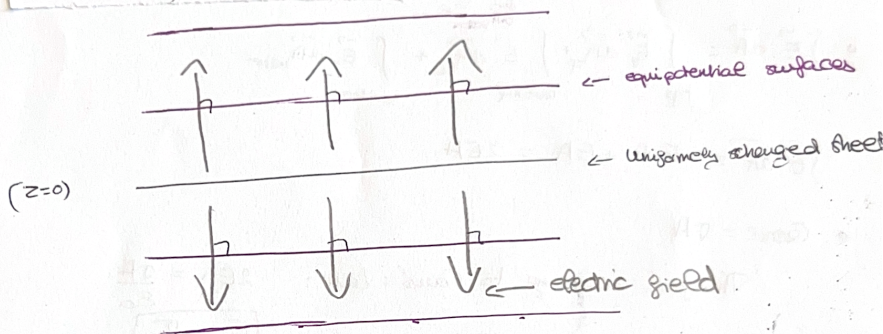
Problem 2

a - $V(z=0) = 0 \Rightarrow V(0) = -\frac{\sigma}{2\epsilon_0}|0| + \text{const} = 0$
 $\Rightarrow \text{constante} = 0$

$$V(x, y, z) = -\frac{\sigma}{2\epsilon_0}|z|$$



b -



- Equipotential surfaces correspond to constant V so constant $|z|$. Therefore we have that equipotentials are planes parallel to the sheet.
- Electric-field lines are orthogonal to equipotentials. The electric field points away from the sheet on both sides (since $\sigma > 0$).

$$E = E_z = -\frac{dV}{dz} \quad : \quad \text{For } z > 0 : E_z = \frac{\sigma}{2\epsilon_0}$$

$$\quad \quad \quad \text{For } z < 0 : E_z = -\frac{\sigma}{2\epsilon_0}$$

The electric field is constant in magnitude

c) V depends only on z

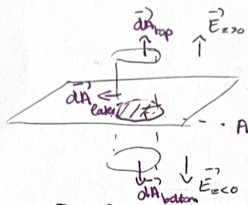
$$\vec{E} = -\frac{\partial V}{\partial z} \hat{z}$$

$$\frac{\partial V}{\partial z} = \begin{cases} -\frac{\sigma}{2\epsilon_0} & (z > 0) \\ +\frac{\sigma}{2\epsilon_0} & (z < 0) \end{cases}$$

$$\text{Therefore } \vec{E}(z) = \begin{cases} \frac{\sigma}{2\epsilon_0} \hat{z} & (z > 0) \\ -\frac{\sigma}{2\epsilon_0} \hat{z} & (z < 0) \end{cases}$$

Magnitude: $E = \frac{\sigma}{2\epsilon_0}$
Direction: outward from the sheet

d)



$$\int \vec{E} \cdot d\vec{A} = \frac{Q_{\text{enc}}}{\epsilon_0}$$

$$\bullet \int \vec{E} \cdot d\vec{A} = \int_{\text{top}} \vec{E}_z d\vec{A}_{\text{top}} + \int_{\text{lateral}} \vec{E} \cdot d\vec{A}_{\text{lateral}} + \int_{\text{bottom}} \vec{E} \cdot d\vec{A}_{\text{bottom}}$$

$$\int \vec{E} \cdot d\vec{A} = EA + EA = 2EA$$

$$\bullet Q_{\text{enc}} = \sigma A$$

$$\text{Therefore, according to Gauss's Law: } 2EA = \frac{\sigma A}{\epsilon_0}$$

$$\boxed{E = \frac{\sigma}{2\epsilon_0}}$$

The direction is outward from the sheet

We find the same result we found in (c)

Problem 2 (20 points)

- a) - Correctly sets constant to ensure $V(0) = 0$ 2 pts
- Correctly plots V vs z 2 pts
- b) - Identifies that equipotentials are planes // to the sheet 2 pts
- Sketches at least two equipotential surfaces clearly 1 pt
- Identifies that electric-field lines are $\perp$ to equipotentials 2 pts
- Correct direction $\vec{E}$ (outward from sheet since $\sigma > 0$) 2 pts
- Correctly states that the electric field is constant in magnitude 1 pt
- (a) c) - Uses $\vec{E} = -\nabla V$ properly 2 pts
- Correctly computes derivative of piecewise function 2 pts
- Correct expression 1 pt
- d) - Correct identification of Gauss's Law 2 pt
- Chooses appropriate Gaussian surface 1 pt
- Correct set up identifying $\oint \vec{E} \cdot d\vec{A} = 2EA$ 2.5 pt
and $Q_{enc} = \sigma A$
- Correctly computes $\vec{E}_{z>0} = \frac{\sigma}{2\epsilon_0} \hat{z}$ and $\vec{E}_{z<0} = -\frac{\sigma}{2\epsilon_0} \hat{z}$ 2.5 pt
- Clearly compares to result in (c) 1 pt

7B – Lecture 001 – Final Exam – Question 3

Rubric and Answer Key

Spring 2025

Problem 3 - 25 Points

This problem is split into four sections, (a), (b), (c), and (d), which I further split into sub-parts for the purpose of partial credit.

Part 1 A cylindrical resistor made of a material of resistivity ρ , with inner radius r_1 , outer radius r_2 and length L , is connected to an ideal battery supplying an emf $\mathcal{E}$. The resistor is made of a conducting material of density d , atomic mass M_a , and has 2 free electrons per atom.

(a) (3pts) Determine the resistance of the cylindrical resistor.

Sub-part 1. (3pts) this part is directly from the formula sheet. integrate for R .

$$R = \int \rho \frac{dl}{A} = \int \rho \frac{dl}{\pi(r_2^2 - r_1^2)} = \rho \frac{L}{\pi(r_2^2 - r_1^2)}$$

(b) (9pts) Determine the drift velocity of the electrons in the resistor. *Hint: it might be helpful to determine the current density throughout the resistor.*

Sub-part 1. (3pts) determine the current density.

$$I = \frac{\mathcal{E}}{R} = \frac{\pi \mathcal{E}(r_2^2 - r_1^2)}{\rho L}$$

Assuming uniform current,

$$J = \frac{I}{A} = \frac{\mathcal{E}}{\rho L}$$

Sub-part 2. (3pts) and electron number density.

$$n = 2 \frac{\text{mass density } d}{\text{mass per atom } M_a} = 2 \frac{d}{M_a}$$

Note the factor of 2 from the number of electrons per atom.

Sub-part 3. (3pts) and drift velocity.

$$|J| = |nQv_d|$$

$$\frac{\mathcal{E}}{\rho L} = 2ev_d \frac{d}{M_a}$$

$$v_d = \frac{\mathcal{E}}{2e\rho} \frac{M_a}{Ld}$$

Part 2 Consider the following circuit, containing two batteries ($V_1 = V_2 = V_0$), four resistors ($R_1 = R_2 = R_3 = R_4 = R$) and one capacitor (C). The switch is initially open, and the capacitor uncharged. At time $t = 0$, the switch is closed. *Express your answers in terms of V_0 , R and C .*

(c) (6pts) Determine the equivalent resistance and equivalent *emf* of the circuit.

Sub-part 1. (4pts) resistance

$$R_{eq} = R_4 + \frac{1}{\frac{1}{R_2} + \frac{1}{R_1 + R_3}} = R + \frac{1}{\frac{1}{R} + \frac{1}{2R}} = \frac{5R}{3}$$

Sub-part 2. (2pts) and *emf*.

$$\mathcal{E} = V_1 + V_2 = 2V_0$$

(d) (7pts) Determine at what time the current reaches 50% of its maximum value.

Sub-part 1. (4pts) current over time.

$$Q(t) = C\mathcal{E} \left(1 - e^{-\frac{t}{ReqC}}\right) = 2CV_0 \left(1 - e^{-\frac{3t}{5RC}}\right) \quad (\text{from formula sheet})$$

$$I(t) = \frac{dQ(t)}{dt} = \frac{6V_0}{5R} e^{-\frac{3t}{5RC}}$$

Sub-part 2. (3pts) final answer.

$$I_{max} = I(0) = \frac{6V_0}{5R}$$

$$\frac{I(t_{50})}{I_{max}} = e^{-\frac{3t_{50}}{5RC}} = 0.5$$

so

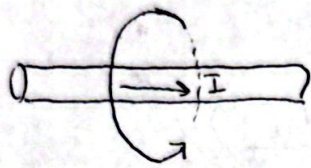
$$t_{50} = \frac{5RC}{3} \ln 2$$

Problem 4

Time: 8mm

$$j(r) = j_0 \frac{a}{r}$$

a)



right hand rule

use Ampere's law b/c symmetry

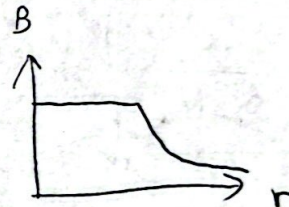
b) $\oint \mathbf{B} \cdot d\mathbf{l} = \mu_0 I_{enc}$

$$r < a$$

$$I_{enc} = \int j(r) dA = \int_0^r j_0 \frac{a}{r} r dr \int_0^{2\pi} d\theta = 2\pi a j_0 r$$

$$B 2\pi r = \mu_0 2\pi a j_0$$

$$\boxed{B = \mu_0 a j_0}$$



$$r > a$$

$$I_{enc} = \int_0^a j(r) dA = 2\pi a^2 j_0$$

$$B 2\pi r = \mu_0 2\pi a^2 j_0 \Rightarrow \boxed{B = \frac{\mu_0 a^2 j_0}{r}}$$

c) $\boxed{I = 2\pi a^2 j_0}$ same as part b?

d) from straight part

$$\boxed{\vec{B} = \frac{\mu_0 I}{2\pi R} \hat{y}}$$

Ampere's law

from circle part

$$B = \frac{\mu_0}{4\pi} \int \frac{I d\mathbf{l} \times \mathbf{r}'}{r'^3} = \frac{\mu_0 I}{4\pi} \int \frac{d\mathbf{l}}{R^2} = \frac{\mu_0 I \cdot 2\pi R}{4\pi R^2} = \boxed{\frac{\mu_0 I}{2R} (-\hat{z})}$$

$$\boxed{\vec{B}_{net} = \frac{\mu_0 I}{2\pi R} \hat{y} - \frac{\mu_0 I}{2R} \hat{z}}$$

Problem 5

Time: 20 min

a) $\vec{J} = nQ\vec{v}$
 $\vec{J} = \oint \vec{J} \cdot d\vec{A} = nQv dA = \frac{Q}{L} \cdot \omega r \cdot A$
 $\vec{J} = dQ\omega$

$$v = \frac{2\pi r}{T} \quad \omega = \frac{2\pi}{T}$$

are you allowed to use dQ ?

$$dI = \frac{dQ \cdot v}{L} = \frac{dQ \omega r}{2\pi r} = \frac{dQ \omega}{2\pi} = \frac{Q \omega dr}{2\pi L} \quad \text{in direction of rotation}$$

$$dQ = \frac{Q}{L} dr$$

is current a vector?

b) $\mu = IA$

$$\mu = \int \pi r^2 \frac{dQ \omega}{2\pi L} = \frac{Q \omega}{2L} \int_0^L r^2 dr = \frac{Q \omega}{2L} \frac{L^3}{3} = \frac{Q \omega L^2}{6} \quad \text{in } \hat{z}$$

c) field of closed loop

$$B = \frac{\mu_0}{4\pi} \int \frac{I d\vec{r} \times \vec{r}'}{r'^3}$$

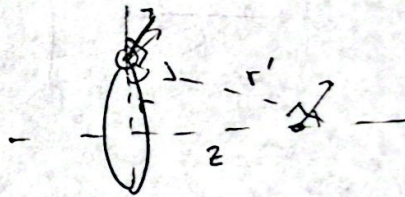
$$= \frac{\mu_0 I}{4\pi} \int \frac{dl}{(r^2 + z^2)} \cdot \frac{r}{\sqrt{r^2 + z^2}} = \frac{\mu_0 I \cdot 2\pi r^2}{4\pi (r^2 + z^2)^{3/2}} = \frac{\mu_0 I r^2}{2(r^2 + z^2)^{3/2}}$$

$$B = \frac{\mu_0 I r^2}{2z^3}$$

for $z \gg r$ $B \sim \frac{\mu_0 I r^2}{2z^3} = \frac{\mu_0 I \pi r^2}{2\pi z^3} = \frac{\mu_0 \mu}{2\pi z^3}$

d)

$$B \sim \frac{\mu_0 Q \omega L^2}{12\pi z^3}$$



7B – Lecture 001 – Final Exam – Question 6

Rubric and Answer Key

Spring 2025

Problem 6 - 25 Points

This problem is split into four sections, (a), (b), (c), and (d), which I further split into sub-parts for the purpose of partial credit.

Three fixed conducting rails of negligible resistance are placed on the sides and at the top of a ramp. The ramp has a total height H , width w , and forms an angle θ with the horizontal direction. A conducting rod of mass m and resistance R is held by an experimentalist and slides down the ramp with constant speed v , as shown in the figure. A uniform magnetic field of magnitude B is applied vertically upward over the entire ramp. The moving rod is initially at the top of the ramp and starts sliding down at time $t = 0$. *You may neglect the effect of the induced magnetic field on the moving rod.*

(a) (6pts) Determine the flux of magnetic field through the conducting loop as a function of time.

Sub-part 1. (3pts) flux versus displacement x of rod down the ramp.

$$\Phi_B = \int \mathbf{B} \cdot d\mathbf{A} = \int_0^w \int_0^x B \cos(\theta) dx' dz' = Bwx \cos(\theta)$$

Sub-part 2. (3pts) flux versus time.

$$x = vt$$

so

$$\Phi_B(t) = Bwvt \cos(\theta)$$

(b) (6pts) Determine the absolute value of the induced current that is generated in the conducting loop, and indicate its orientation on a diagram.

Sub-part 1. (3pts) induced current.

$$\mathcal{E} = -\frac{d\Phi_B}{dt} = -Bwv \cos(\theta)$$

$$|I| = \left| \frac{\mathcal{E}}{R} \right| = \frac{Bwv \cos(\theta)}{R}$$

Sub-part 2. (3pts) nature abhors a change in flux.

Remark. Nature abhors a change in flux.

NATURE ABHORS A CHANGE IN FLUX

Nature abhors a change in flux.

David J. Griffiths

The flux is upwards through the loop and increasing, so the induced current will want to generate a field that opposes that (downwards). The induced current will be clockwise when viewed from the top (see Fig. 1).

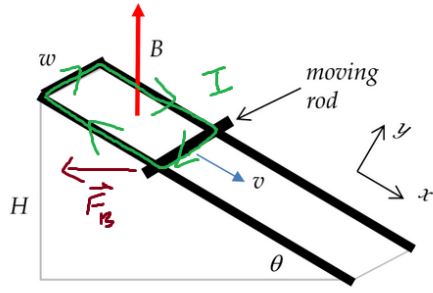


Figure 1: Diagram for Problem 6 (b) and (c)

- (c) (6pts) Determine the magnitude of the magnetic force acting on the moving rod and indicate its direction on a diagram.

Sub-part 1. (3pts) magnitude.

$$d\mathbf{F}_m = I d\mathbf{l} \times \mathbf{B}$$

as $d\mathbf{l} \times \mathbf{B}$ is constant over the length w of the rod, this can be integrated to

$$\mathbf{F}_m = I \mathbf{w} \times \mathbf{B}$$

As $\mathbf{w}$ and $\mathbf{B}$ are orthogonal,

$$|F_m| = IwB = \frac{B^2 w^2 v \cos(\theta)}{R}$$

Sub-part 2. (3pts) and direction.

The magnetic force will point horizontally to the left. Note this is not in the direction of the ramp, but the direction perpendicular to both $\mathbf{B}$ and the rod (see Fig. 1).

Now instead of being maintained at constant speed, the rod is released from the top and slides down without friction. After speeding up, the rod reaches a constant speed called terminal speed.

- (d) (7pts) Determine the terminal speed of the rod.

Sub-part 1. (4pts) net force.

We need to consider the net force in the $\hat{x}$ direction, where $\hat{x}$ is defined to be the direction parallel to the slope of the ramp. (This is because the normal force will counteract any component of the force in the $\hat{y}$ direction).

$$\mathbf{F}_m \cdot \hat{x} = -\frac{B^2 w^2 v \cos(\theta)}{R} \cos(\theta)$$

$$\mathbf{F}_g \cdot \hat{x} = mg \sin(\theta)$$

$$\mathbf{F}_{net} \cdot \hat{x} = mg \sin(\theta) - \frac{B^2 w^2 v \cos^2(\theta)}{R}$$

Sub-part 2. (3pts) terminal speed.

The terminal speed v_T will be the speed at which $\mathbf{F}_{net} \cdot \hat{x} = 0$.

$$v_T = \frac{mgR \sin(\theta)}{B^2 w^2 \cos^2(\theta)}$$

Sanity checks! When $\theta = 0$, the ramp is horizontal so the terminal velocity is zero. When $\theta = \pi/2$, the ramp is vertical so there is no terminal speed – the rod will keep falling faster and faster. This is because of two effects: there will be no magnetic flux as the loop is going with the $\mathbf{B}$ field, so there will be no current. Also, even if there was current, it would be into the ramp so it would not affect the falling.